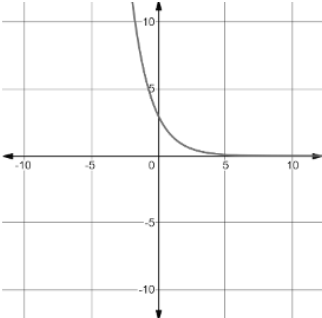
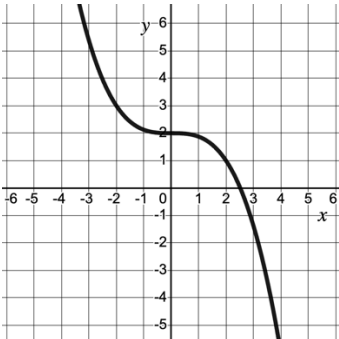


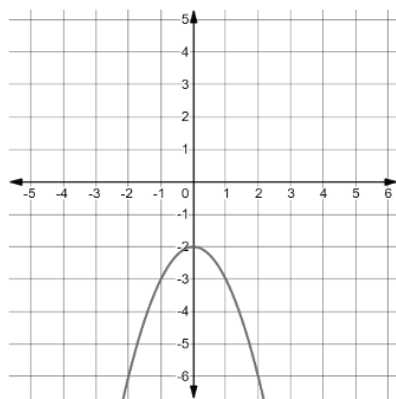
Algebra 1
Unit 13
Lesson 4 Practice

Name Answer Key
Date _____
Period _____

Determine which family of functions each transformation is part of.

Representation of Transformation		Equation of Parent Function													
1.	<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>-9</td></tr><tr><td>-1</td><td>-11</td></tr><tr><td>0</td><td>-13</td></tr><tr><td>1</td><td>-15</td></tr><tr><td>2</td><td>-17</td></tr></table>	x	y	-2	-9	-1	-11	0	-13	1	-15	2	-17	☒ $y = x$ ☐ $y = x^2$ ☐ $y = x^3$ ☐ $y = \sqrt{x}$	☐ $y = x$ ☐ $y = 2^x$ ☐ $y = \frac{1}{2}^x$
x	y														
-2	-9														
-1	-11														
0	-13														
1	-15														
2	-17														
2.		☐ $y = x$ ☐ $y = x^2$ ☐ $y = x^3$ ☐ $y = \sqrt{x}$	☐ $y = x$ ☐ $y = 2^x$ ☒ $y = \frac{1}{2}^x$												
3.	<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>-12</td></tr><tr><td>-1</td><td>-3</td></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>-3</td></tr><tr><td>2</td><td>-12</td></tr></table>	x	y	-2	-12	-1	-3	0	0	1	-3	2	-12	☐ $y = x$ ☒ $y = x^2$ ☐ $y = x^3$ ☐ $y = \sqrt{x}$	☐ $y = x$ ☐ $y = 2^x$ ☐ $y = \frac{1}{2}^x$
x	y														
-2	-12														
-1	-3														
0	0														
1	-3														
2	-12														
4.	$y = \frac{2}{3}x + 6$	☒ $y = x$ ☐ $y = x^2$ ☐ $y = x^3$ ☐ $y = \sqrt{x}$	☐ $y = x$ ☐ $y = 2^x$ ☐ $y = \frac{1}{2}^x$												
5.		☐ $y = x$ ☐ $y = x^2$ ☒ $y = x^3$ ☐ $y = \sqrt{x}$	☐ $y = x$ ☐ $y = 2^x$ ☐ $y = \frac{1}{2}^x$												

6.



☐ $y = x$
☐ $y = x^2$
☒ $y = x^3$
☐ $y = \sqrt{x}$

☐ $y = |x|$
☐ $y = 2^x$
☐ $y = \frac{1^x}{2}$

7.

x	-2	-1	0	1	2
y	18	6	2	$\frac{2}{3}$	$\frac{2}{9}$

☐ $y = x$
☐ $y = x^2$
☐ $y = x^3$
☐ $y = \sqrt{x}$

☐ $y = |x|$
☐ $y = 2^x$
☒ $y = \frac{1^x}{2}$

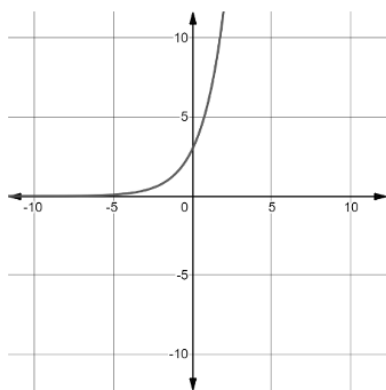
8.

$$y = \sqrt{x+6} - 4$$

☐ $y = x$
☐ $y = x^2$
☐ $y = x^3$
☒ $y = \sqrt{x}$

☐ $y = |x|$
☐ $y = 2^x$
☐ $y = \frac{1^x}{2}$

9.



☐ $y = x$
☐ $y = x^2$
☐ $y = x^3$
☐ $y = \sqrt{x}$

☐ $y = |x|$
☒ $y = 2^x$
☐ $y = \frac{1^x}{2}$

10.

$$y = 2|x - 5|$$

☐ $y = x$
☐ $y = x^2$
☐ $y = x^3$
☐ $y = \sqrt{x}$

☒ $y = |x|$
☐ $y = 2^x$
☐ $y = \frac{1^x}{2}$